Varnish: A Performance Booster for Web Applications

Here's a superior caching engine for your Web applications. It will get them to work at blazing speeds with minimal configuration.

Web applications have evolved immensely and are capable of doing almost everything you would expect from a native desktop application. With this evolution, the amount of data and the accompanying need for processing has also increased. Apparently, a full-fledged Web application in a production set-up needs high end infrastructure and adds a lot of latency on the server side for processing repetitive jobs by different users. Varnish is a super-fast caching engine, which can reside in front of any Web server to cache these repeated requests and serve them instantly.

Why Varnish?

Varnish has several advantages over other caching engines. It is lightweight, easy to set up, starts working immediately, works independently with any kind of backend Web server and is free to use (FreeBSD licence). Varnish is highly customisable, for which the Varnish Configuration Language (VCL) is used.

The advantages of this tool are:

- Lightweight, easy to set up, good documentation and forum support
- Zero downtime on configuration changes (always up)
- Works independently with any Web server and allows multi-site set up with a single Varnish instance
- Highly customisable with an easy configuration syntax
- Admin dashboard and other utilities for logging and performance evaluation
- Syntax testing and error detection of configuration without activation

There are only a few limitations to this tool. Varnish does not support the HTTPS protocol, but it can be configured as an HTTP reverse proxy using Pound for internal caching. Also, the syntax of VCL has been changing for various commonly used configurations with the newer versions of Varnish. It is recommended that users refer to the documentation for the exact version to avoid mistakes.

Varnish is a powerful tool and allows you to do a lot more. For instance, it can be used to give temporary 301 redirections or serve your site while the backend server is down for maintenance.

Configuration and usage

Let us go through the steps to install and configure Varnish. For this tutorial, we'll use Ubuntu 14.04 LTS with the NGINX server.

Varnish 4.1 is the latest stable release, which is not available in Ubuntu's default repositories. Hence, we need to add the repository and install Varnish using the following commands:

```
apt-get install apt-transport-https
curl https://repo.varnish-cache.org/GPG-key.txt | apt-key
add -
echo "deb https://repo.varnish-cache.org/ubuntu/ trusty
varnish-4.1" \
```